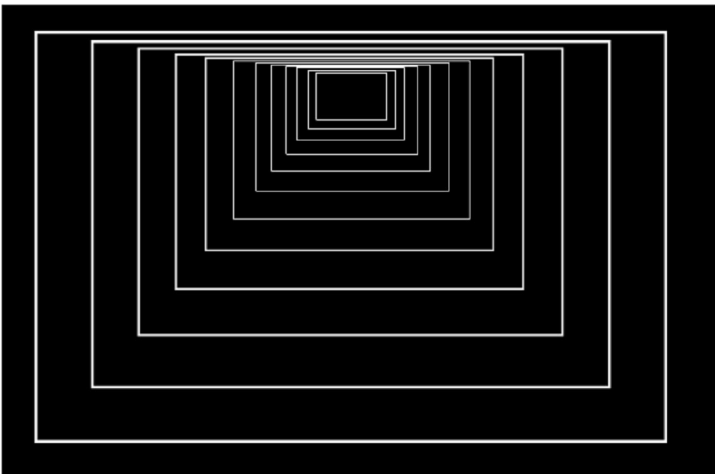


Getting Recursive with Bench Calculator

This fourth article on the mathematical journey through open source takes readers through the recursive functional power of the bench calculator.



Equipped with the fundamentals of programming functions, let's jump into the spiral of recursion. Frankly speaking, recursion is not a spiral, as many would like to picture it; it is more to do with the way our brains work. So, let's look into it as our brain suggests. Any recursion fundamentally has two parts: the recursive relation and the termination condition(s). For mathematics, recursion is basically to do with functions.

Recursive relation

A function, when expressed or defined using the function itself, but of a lower order, is called a recursive relation. Typical lower-order examples would be $f(n)$ expressed in $f(n-1)$, $f(n-2)$, ... where n is an integer; and $f(x)$ expressed in $f(x/2)$, ... where x is a real number. In case of mathematics, a recursive relation is also called a recurrence relation. Here are some examples:

- Factorial: $n! = n * (n-1)!$ or $\text{fact}(n) = n * \text{fact}(n-1)$
- Fibonacci: $\text{fib}(n) = \text{fib}(n-1) + \text{fib}(n-2)$
- Exponent: $e^x = (e^{x/2})^2$ or $\text{exp}(x) = (\text{exp}(x/2))^2$

Termination condition

Now, if you try to expand these, they will continue forever, unless you decide to stop at some decent condition. For example, $n! = n * (n-1)! = n * (n-1) * (n-2)! = n * (n-1) * (n-2) * (n-3)! = \dots$ and so on. So, let's decide to stop it when you reach 0, assuming n to be positive. And hence, $n!$ becomes $n * (n-1) * (n-2) * \dots * 2 * 1 * 0!$. And then you can define $0!$ to be 1. This is called the termination condition. And you would notice that, whenever there is a recursive relation, you would need one or more such termination conditions to stop the relation going on forever. The number of termination conditions typically depends on the different lower orders you have in the recursive relation. For example, the factorial of n has just one lower-order factorial of $(n-1)$; the exponent of x again has just one lower-order exponent of $(x/2)$; but Fibonacci of n has two lower orders: Fibonacci of $(n-1)$ and Fibonacci of $(n-2)$. Hence, the factorial and exponent would have one termination condition, and Fibonacci would have two termination conditions.